

Introduction to Quantum Computing

Last Lab Exercise: QISKIT

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1. Run the adjoint Jupyter Notebook *Quantum Teleportation.ipynb* and discuss the differences between the simulated results (under the assumption of a perfect quantum computer) and the real quantum computers from IBM Quantum Platform.

Be careful, because the cell

```
# INCLUDE YOUR 'jobid' BELOW TO LOAD YOUR DATA;  
# RUN THIS CELL FOR RETRIEVING FROM IBM QUANTUM PLATFORM OR FROM LOCAL FILE  
job_id = 'jobid'
```

should be modified so that the `job_id` variable points to the *job-id* generated by the IBM Quantum Platform and printed right above such cell. Note that for retrieving the results you only need to run the code *from* the definition of this cell. It is NOT necessary to run all the simulation and to resubmit another job to the IBM Quantum Platform. The results are also saved to file.

2. Using the adjoint Notebook as a template, implement a quantum circuit to sum two 4-qubit numbers $|a\rangle$ and $|b\rangle$ via the QFT and IQFT circuits. You can use the QISKIT library¹ or implement your own QFT and IQFT operators. Initialise the two quantum registers $|a\rangle \times |b\rangle$ to the entangled state:

$$\frac{1}{\sqrt{2}} (|2\rangle \times |9\rangle + |6\rangle \times |5\rangle)$$

Draw the circuit on the Jupyter Notebook

3. Simulate the circuit on *qiskit-aer* using a simulator of a perfect quantum computer. Afterwards, compare the result with a real quantum computer from IBM Quantum Platform. Discuss your results.

You should upload to Moodle a compressed ZIP (or 7-ZIP) file both Jupyter Notebooks (with their results), the `result_*.json` file and a PDF with your discussions.

¹See: <https://docs.quantum.ibm.com/api/qiskit/qiskit.circuit.library.QFT>